

EDMONTON BLATCHFORD, AB, Canada

WMO: 711570

Lat: 53.5799N

Lon: 113.5156W

Elev: 671

StdP: 93.52

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.4	(c) -25.4	(d) -33.0	(e) 0.2	(f) -27.9	(g) -29.7	(h) 0.3	(i) -25.0	(j) 10.5	(k) -4.4	(l) 9.3	(m) -5.5	(n) 2.7	(o) 210	(p) 0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.1	(c) 28.4	(d) 17.7	(e) 26.7	(f) 16.9	(g) 25.1	(h) 16.2	(i) 19.4	(j) 26.2	(k) 18.2	(l) 24.6	(m) 17.2	(n) 23.2	(o) 4.1	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.0	13.2	22.4	15.8	12.2	21.1	14.7	11.3	20.1	58.2	26.3	54.4	24.5	51.0	23.2	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.9	(b) 8.5	(c) 7.5	(e) -31.2	(f) 31.9	(g) 3.5	(h) 1.9	(i) -33.7	(j) 33.2	(k) -35.7	(l) 34.4	(m) -37.6	(n) 35.4	(o) -40.2	(p) 36.8		
(a) 9.9	(b) 8.5	(c) 7.5	(e) -31.3	(f) 21.3	(g) 3.4	(h) 1.4	(i) -33.8	(j) 22.3	(k) -35.8	(l) 23.1	(m) -37.7	(n) 23.8	(o) -40.2	(p) 24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-10.1	-8.2	-3.1	4.8	11.5	15.6	18.2	17.0	12.0	5.2	-3.4	-9.1		
	DBStd	11.78	8.68	7.51	7.48	5.39	4.68	3.26	3.12	3.34	4.38	4.72	6.68	7.69		
	HDD10.0	2926	624	508	408	167	38	2	0	1	28	159	401	591		
	HDD18.3	5209	883	742	666	405	214	93	42	66	192	407	651	849		
	CDD10.0	836	0	0	0	12	85	170	253	217	89	10	0	0		
	CDD18.3	77	0	0	0	0	3	11	37	23	3	0	0	0		
	CDH23.3	781	0	0	0	2	60	112	321	224	59	2	0	0		
	CDH26.7	157	0	0	0	0	9	18	74	45	11	0	0	0		
	WSAvg	3.4	3.0	3.0	3.4	4.0	4.0	3.8	3.6	3.3	3.4	3.5	3.1	3.0		
	PrecAvg	481	25	18	18	23	47	81	95	68	44	19	17	23		
	PrecMax	626	73	46	52	54	142	196	185	169	115	66	51	54		
(15) Precipitation	PrecMin	337	2	2	4	1	4	13	30	8	3	0	0	3		
	PrecStd	75	18	11	12	12	27	45	43	42	26	15	13	13		
	DB	7.3	8.3	13.6	22.7	28.0	28.9	31.3	30.3	28.1	21.9	13.0	8.3			
	MCWB	3.0	3.4	6.5	11.3	14.3	17.0	19.3	18.7	16.6	12.6	6.7	3.4			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	5.2	5.4	10.1	19.1	24.9	26.2	28.6	27.7	24.8	17.3	8.8	5.2			
	2% MCWB	2.0	1.6	4.3	9.3	13.2	15.5	18.7	17.9	15.1	10.0	4.5	1.8			
	5% DB	3.4	3.5	7.9	16.1	22.6	24.3	26.7	25.8	22.4	14.4	6.5	3.2			
	5% MCWB	0.9	0.5	3.3	7.8	12.1	14.7	17.8	17.2	14.2	8.3	2.9	0.5			
	10% DB	1.3	1.7	5.8	13.5	20.1	22.3	24.9	24.0	19.9	12.1	4.5	1.2			
	10% MCWB	-0.8	-0.7	1.9	6.3	10.9	14.1	17.0	16.5	12.9	6.9	1.7	-1.1			
	0.4% WB	3.6	3.8	7.1	11.7	15.4	18.8	21.2	20.5	17.7	13.0	7.1	3.9			
	0.4% MCDB	6.5	7.8	12.5	21.5	24.4	26.1	27.7	26.9	25.6	20.3	11.8	7.7			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	2.2	2.0	4.9	9.8	13.9	16.9	19.9	19.0	15.9	10.5	4.9	2.0			
	2% MCDB	4.8	4.8	9.5	18.2	22.3	23.3	26.5	25.9	23.4	16.3	8.3	4.8			
	5% WB	1.0	0.7	3.5	8.2	12.7	15.8	18.7	18.0	14.6	8.8	3.3	0.6			
	5% MCDB	3.3	3.2	7.4	15.1	21.0	22.1	25.1	24.3	21.6	13.4	6.0	3.1			
	10% WB	-0.8	-0.7	2.2	6.8	11.6	14.9	17.7	16.9	13.4	7.5	2.0	-1.0			
	10% MCDB	1.2	1.6	5.4	12.6	19.0	20.8	23.5	22.8	18.9	11.3	4.2	1.2			
	MDBR	8.0	8.8	8.7	10.3	11.0	9.9	10.1	10.5	10.7	9.2	7.3	7.7			
	5% DB	9.5	11.0	11.6	14.8	14.5	12.9	12.9	13.6	15.4	13.6	9.7	9.5			
(36) Mean Daily Temperature Range	5% WB	7.3	8.3	7.6	7.9	6.4	5.5	5.6	6.0	7.5	7.6	6.5	7.2			
	MCDBR	9.4	10.7	11.0	14.0	13.1	11.0	11.6	12.2	14.4	12.8	9.3	9.5			
	MCWBR	7.5	8.3	7.4	7.8	6.1	5.3	5.9	6.1	7.2	7.6	6.5	7.6			
	MCWBR	7.5	8.3	7.4	7.8	6.1	5.3	5.9	6.1	7.2	7.6	6.5	7.6			
(40) Clear-Sky Solar Irradiance	taub	0.257	0.267	0.312	0.327	0.343	0.353	0.356	0.362	0.308	0.282	0.254	0.240			
	taud	2.307	2.356	2.308	2.389	2.392	2.380	2.418	2.351	2.499	2.547	2.456	2.315			
	Ebn at Noon	731	851	872	906	905	896	886	857	875	832	748	683			
	Edh at Noon	57	75	99	105	110	113	107	108	82	62	49	46			
(44) All-Sky Solar Radiation	RadAvg	0.97	2.01	3.56	4.67	5.73	6.03	6.16	5.07	3.50	2.01	0.96	0.72			
	RadStd	0.11	0.16	0.27	0.44	0.42	0.39	0.30	0.35	0.36	0.24	0.11	0.07			

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
(46) Station Only	DBAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)	DBAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page